YOUR TITLE HERE

Authors: MAKE SURE TO WRITE YOUR NAMES

Generative AI (genAI) Usage Statement

Write a short statement about how you used genAI on this assignment. *Note*: If you don't write anything, we'll assume you used genAI to write the whole thing!!!

We did not use generative AI on this assignment.

(Write your response here)

1. Modeling Question

(State your modeling question directly. Categorize it according to the action/specificity framework.)

(You will likely need to write 2-3 paragraphs for this section.)

Suggested template: FOR INSPIRATION, NOT TO USE DIRECTLY.

A: we were curious about some question because of some reasons

B: but we weren't sure how to model it

T: so we did some background research for ideas (i.e. research synthesis matrix)

We were curious about how distance between houses impacts disease transmission, but we weren't sure how exactly to map it out, so we decided to simulate a town, Needham, and did background research.

2. Literature Search

(Identify 3 or more scholarly articles that are relevant to your modeling question. Analyze these references using the synthesis matrix approach. Report your synthesis matrix in a readable format: Include a link to a Google Sheet if it does not fit in the report.)

(You will likely need to write 2-3 paragraphs for this section.)

A: we learned X from article 1, Y from article 2, and Z from article 3

B: but we didn't find a direct answer to our modeling question

T: so we went on to build a model

3. Natural World

(Describe the natural world using your 3 or more scholarly articles. Use an inline citation every time you refer to a fact from one of the scholarly articles. Everything you describe should be relevant to your modeling question. This section should connect to your tenchi diagram; we recommend putting the diagram directly between sections 3 and 4 since it serves as a transition between the two.)

(You will likely need to write 2-3 paragraphs for this section.)

Suggested template: FOR INSPIRATION, NOT TO USE DIRECTLY.

A: here's a few things that are important in the natural world

B: but there are some things we assumed are not relevant to our modeling question

T: so, we left those things outside the system boundary (they are not modeled)

4. Model: Representation & Assumptions

(Describe how your model represents the natural world. Use a tenchi diagram to illustrate what parts of the natural world are inside the system boundary. In the same tenchi diagram, provide a visual of your model. Describe in words how your model represents the chosen system. Describe in words what assumptions you have made. Cite your references when appropriate.)

(You will likely need to write 5-7 code & text regions for this section. You should have images and equations to describe your model, and then implement your model in code.)

Suggested template: FOR INSPIRATION, NOT TO USE DIRECTLY.

A: this part of the natural world is important for our question

B: but we can't include that directly because of X reason

T: so here is the way we represented it / what we assumed to simplify

One assumption that we had was that the 120 house sample of needham that we took was accurate

https://www.aimspress.com/aimspress-data/mbe/2026/7/PDF/mbe-23-07-080.pdf

5. Verification & Validation

(Select appropriate verification facts, and implement them as assert statements. In Projects 2 and 3, carry out model validation.)

(You will likely need to write 2-3 code sections for this section, and use words to describe your argument for verification and validation.)

Suggested template: FOR INSPIRATION, NOT TO USE DIRECTLY.

A: here is a fact that should be true about the model

B: but it is not immediately obvious this is true, because X

T: so we carried out this test to show that it is true

6. Claims & Limitations

(Run simulations using your model, and make claims about the natural world based on those results. Connect each assumption to a limitation about what you cannot *conclude based on your model.)*

(You will likely need to write 2-3 code sections for this section [generating results with your model], and 2-3 paragraphs making claims using your results, and discussing the limitations.)

Suggested template: FOR INSPIRATION, NOT TO USE DIRECTLY.

A: remember this assumption from Section 4

B: but [our model still captures X important trend OR our question is about Y specifically OR our results are still informative because of Z]

T: so, we can conclude X, keeping in mind Y caveats

```
function [Sh, Ih, Rh] = simulate_absir(M, Iv0, T, infection_rate, recovery_rate)
% Simulate agent-based transmission model. Uses a graph to represent social
% connectivity between agents.
%
% Note: You may find it helpful to summarize the state history matrices via
% summation. For instance sum(Ih, 1) will return the total number of
% infected persons at each timestep in the simulation.
%
% Inputs
%   M (square matrix): Adjacency matrix
%   Iv0 (column vector): Initial infection state
```

```
%   T (integer): Number of timesteps to simulate
%   infection_rate (float): Infection rate (probabilistic)
%   recovery_rate (float): Recovery rate (probabilistic)
%
% Returns
%   Sh (matrix): Susceptible state history
%   Ih (matrix): Infected state history
%   Rh (matrix): Recovered state history

% Setup
dim = length(Iv0);  % Dimensions of initial state
Ih = zeros(dim, T); % Infection history
Ih(:, 1) = Iv0; % Record the initial state to infection history
Rh = zeros(dim, T); % Recovery history

% Construct an "action" helper function
    function [I, R] = action(I, R)
        % Compute infection probabilities based on the social graph
        v_eff = infection_rate * M * I;
        v_pr = v_eff ./ (1 + v_eff);

        % Draw random values
        v_infect = rand(dim, 1) <= v_pr;
        v_recover = rand(dim, 1) <= recovery_rate;

        % Infect non-recovered individuals
        I = I | v_infect & (~R);
        % Recover infected individuals
        R = R | (I & v_recover);
        I = I & (~R);
    end

% Run simulation
for i = 2:T
    [Ih(:, i), Rh(:, i)] = action(Ih(:, i-1), Rh(:, i-1));
end

% Compute susceptible history
Sh = ones(dim, T) - Ih - Rh;
end


function strength = distancetostrength(d)
strength = exp(-d/50);
%strength = 1/(1+d);
end

function M_res = make_adj_pods(n_per_pod, n_pods, dBlock)
% Construct an adjacency matrix for a "podding" scenario
%
```

```
% Usage
%   M_res = make_adj_pods(n_per_pod, n_pods, infectivity)
%
% Arguments
%   n_per_pod  = number of persons per "pod"
%   n_pods     = total number of pods; total population = n_per_pod * n_pods
%   pod_effect = relative infectivity within pods; should be infectivity > 1
%
% Returns
%   M_res = adjacency matrix for desired graph
n_total = n_per_pod * n_pods;

% Create the "base matrix" with weak connections between all persons
M_res =  distancetostrength(dBlock)* (ones(n_total, n_total) - eye(n_total));

% Add the pod structure
for i = 1:n_pods
    % Determine the slicing indices
    i0 = (i - 1) * n_per_pod + 1; ie = i0 + n_per_pod - 1;
    % Assign strong in-pod weights
    M_res(i0:ie, i0:ie) = 10 * M_res(i0:ie, i0:ie);
end

end
```

```
% x = zeros(120,1);
% for i = 1:120
%     prob = rand * 12;
%     if prob > 1
%         x(i) = 0;
%     end
%     if prob < 1
%         x(i) = 1;
%     end
% end

initial_state = zeros(120, 1);
initial_state(1:10) = 1;



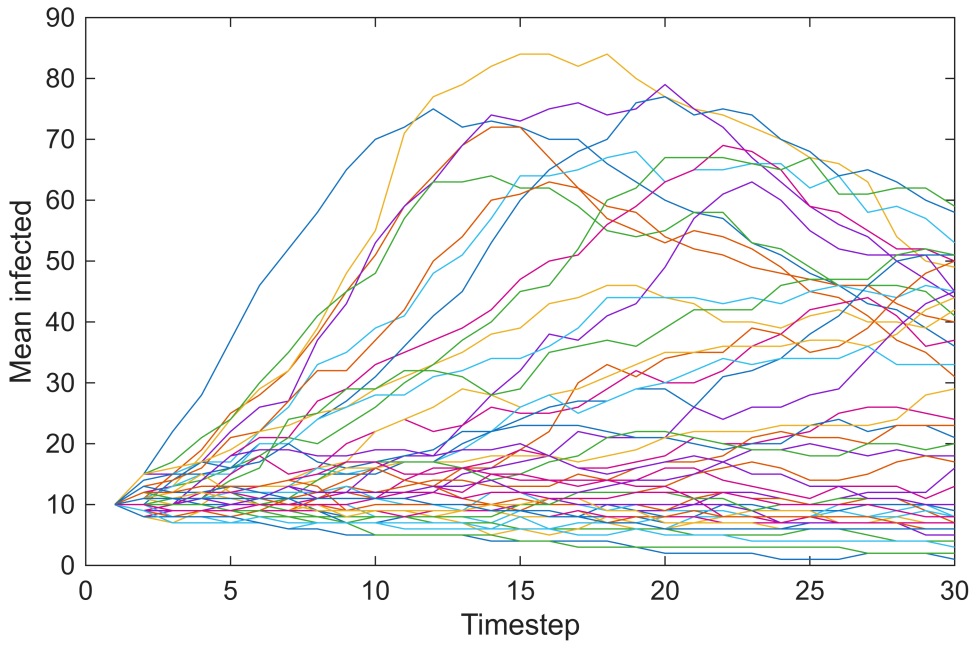
t_end = 30;
infection_rate = 0.005;
recovery_rate = 0.05;
```

```
% number of different distances
num_steps = 50;

% range of possible distances
min_distance = 50; max_distance = 200;

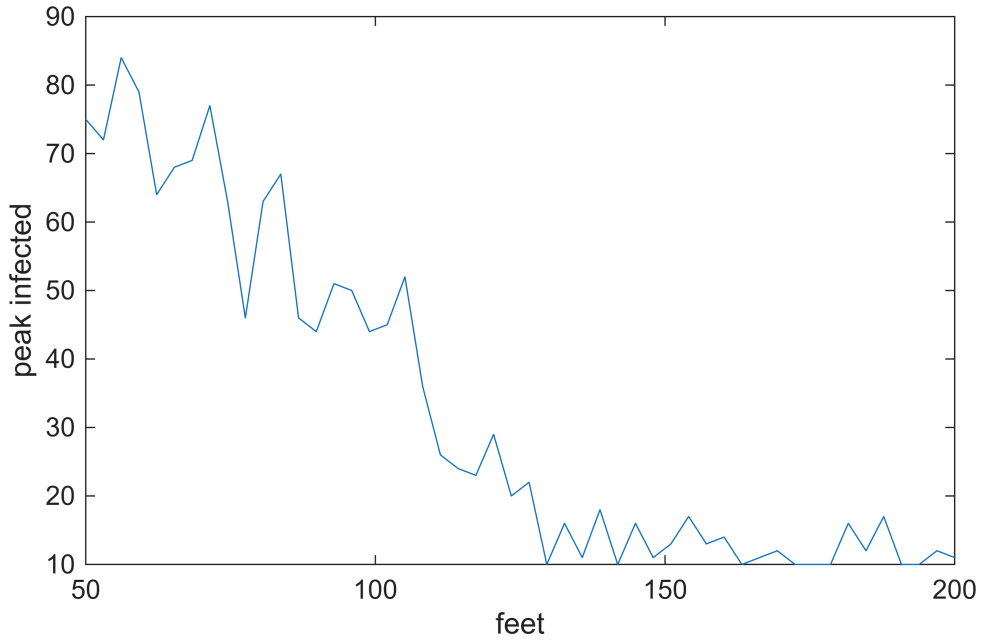
% array of possible distances
distance_vals = linspace(min_distance, max_distance, num_steps);
I_temp = zeros(num_steps, t_end);
% simulate each distance
for k = 1:num_steps
    adj_matrix = make_adj_pods(20, 6, distance_vals(k));
    for t = 1:t_end
        [~, Ih, ~] = simulate_absir(adj_matrix, initial_state, t_end,
infection_rate, recovery_rate);
        I_temp(k,:) = sum(Ih, 1);
    end
end

figure
plot(1:30, I_temp')
xlabel('Timestep'); ylabel('Mean infected')
```




```
peaks = max(I_temp, [], 2);

figure
plot(distance_vals, peaks)
xlabel('feet'); ylabel('peak infected')
```



7. Conclusion

(Restate your modeling question, and refer to the claims from Section 6. Use your claims to directly answer your modeling question.)

(You will likely need just 1 paragraph for this section.)

Suggested template: FOR INSPIRATION, NOT TO USE DIRECTLY.

A: this part of the natural world is important, and X is already known

B: but Y is not yet known, so we built a model to answer the question ...

T: so, given [particular result from Section 6], we conclude Z

8. Future Work

(Based on your limitations from Section 6, state 2 or more ideas of future work for other modelers to attempt.)

(You will likely need 2-3 paragraphs for this section.)

Suggested template: FOR INSPIRATION, NOT TO USE DIRECTLY.

A: our model assumed X

B: but the natural world is actually like Y

T: so there are open questions, such as Z

References

(List all references alphabetically by last name. Use the following format: Last name 1, Last name 2, ... (Year) "Title of Reference")